Math 161
Midterm Exam 2 solutions

Justin Campbell

1. (a) For any $x \neq -2$, we have

$$\frac{x^2 - 3x + 10}{x+2} = \frac{(x+2)(x-5)}{x+2} = x - 5.$$

The limit in question only depends on the values of the function for $x \neq -2$, so

$$\lim_{x \to -2} \frac{x^2 - 3x + 10}{x+2} = \lim_{x \to -2} x - 5.$$

But $x - 5$ is a polynomial function and hence continuous, so

$$\lim_{x \to -2} x - 5 = -2 - 5 = -7.$$

(b) First we use the fact that

$$\lim_{x \to \infty} f(x) = \lim_{x \to 0^+} f(\tfrac{1}{x}),$$

which was exercise 3(a) on Problem Set 5. This implies that

$$\lim_{x \to \infty} \frac{1 - x^2}{2x - 3} = \lim_{x \to 0^+} \frac{1 - \frac{1}{x^2}}{\frac{2}{x} - 3}.$$

For any $x \neq 0$, we have

$$\frac{1 - \frac{1}{x^2}}{\frac{2}{x} - 3} = \frac{x^2 - 1}{2x - 3x^2},$$

and hence

$$\lim_{x \to 0^+} \frac{1 - \frac{1}{x^2}}{\frac{2}{x} - 3} = \lim_{x \to 0^+} \frac{x^2 - 1}{2x - 3x^2},$$

which we claim does not exist. If it did, then by the theorem on products of limits, we would have

$$\begin{aligned} \lim_{x \to 0^+} \frac{x^2 - 1}{2 - 3x} &= \lim_{x \to 0^+} x \cdot \frac{x^2 - 1}{2x - 3x^2} \\ &= \left(\lim_{x \to 0^+} x \right) \left(\lim_{x \to 0^+} \frac{x^2 - 1}{2x - 3x^2} \right) \\ &= 0 \cdot \left(\lim_{x \to 0^+} \frac{x^2 - 1}{2x - 3x^2} \right) = 0. \end{aligned}$$

But

$$\lim_{x \to 0^+} \frac{x^2 - 1}{2 - 3x} = \frac{0^2 - 1}{2 - 3 \cdot 0} = -\frac{1}{2}$$

because $\frac{x^2-1}{2-3x}$ is a rational function its denominator does not vanish at 0, hence is continuous there. This is a contradiction.

(c) Proceeding as in part (b), we obtain

$$\lim_{x\to\infty} \frac{x-1}{x^2+1} = \lim_{x\to 0^+} \frac{\frac{1}{x}-1}{\frac{1}{x^2}+1} = \lim_{x\to 0^+} \frac{x-x^2}{1+x^2}.$$

Since $\frac{x-x^2}{1+x^2}$ is a rational function and its denominator does not vanish at 0, it is continuous at 0, so

$$\lim_{x\to 0^+} \frac{x-x^2}{1+x^2} = \frac{0-0^2}{1+0^2} = 0.$$

2. We are assuming that $\lim_{x\to 0} f(x) = f(0)$, and we want to show that $\lim_{x\to a} f(x) = f(a)$ for any $a \in \mathbb{R}$. By exercise 1 on Problem Set 4, we have

$$\lim_{x\to a} f(x) = \lim_{h\to 0} f(a+h).$$

Then, using the equation $f(x+y) = f(x) + f(y)$ and the theorem on sums of limits, we have

$$\lim_{h\to 0} f(a+h) = \lim_{h\to 0} (f(a) + f(h)) = \lim_{h\to 0} f(a) + \lim_{h\to 0} f(h) = f(a) + f(0).$$

But

$$f(0) = f(0+0) = f(0) + f(0)$$

implies that $f(0)$, so the previous line says that

$$\lim_{h\to 0} f(a+h) = f(a) + f(0) = f(a)$$

as needed.

Alternatively, here is a more direct ϵ-δ argument. We want to show that $\lim_{x\to a} f(x) = f(a)$ for any $a \in \mathbb{R}$, i.e. for any $\epsilon > 0$, there exists $\delta > 0$ such that for any $x \in \mathbb{R}$, the inequality $|x-a| < \delta$ implies that $|f(x) - f(a)| < \epsilon$. So fix $\epsilon > 0$. We are assuming that $\lim_{x\to 0} f(x) = f(0)$, which means that there exists $\delta > 0$ such that for any $x' \in \mathbb{R}$, the inequality $|x'| < \delta$ implies that $|f(x')| < \epsilon$. In particular, we can take $x' = x - a$, whence $|x-a| < \delta$ implies that $|f(x-a)| < \epsilon$. But we have

$$f(x) - f(a) = f(x-a+a) - f(a) = f(x-a) + f(a) - f(a) = f(x-a),$$

from whence we obtain

$$|f(x) - f(a)| = |f(x-a)| < \epsilon.$$

3. Let $g : \mathbb{R} \to \mathbb{R}$ be the function defined by $g(x) := f(x) - x^2$. Note that g is a difference of continuous functions and hence continuous. We also observe that

$$g(0) = f(0) - 0^2 = 3 \text{ and } g(2) = f(2) - 2^2 = -3.$$

In particular $g(2) < 0 < g(0)$, so by the intermediate value theorem there exists $x \in [0,2]$ such that $g(x) = 0$. This means that for this value of x we have

$$f(x) - x^2 = g(x) = 0,$$

i.e. $f(x) = x^2$.